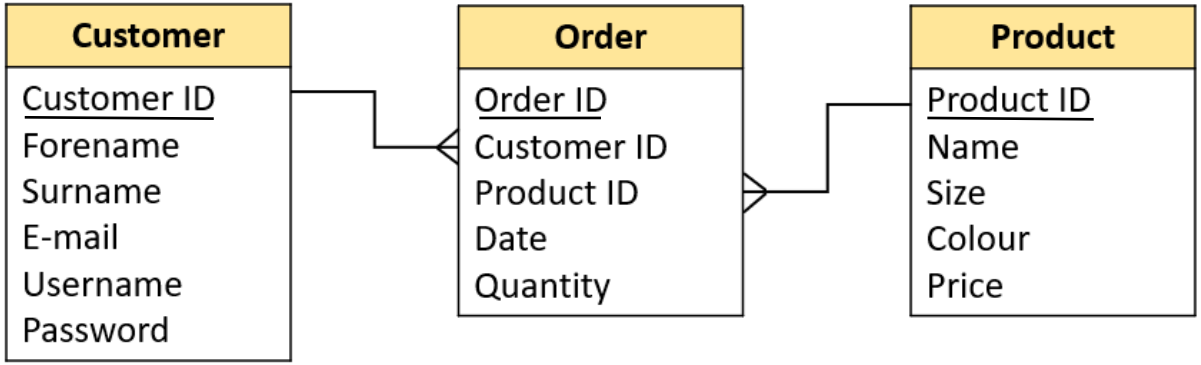
SQL Design Examples & Exercises

1.) Creating Tables



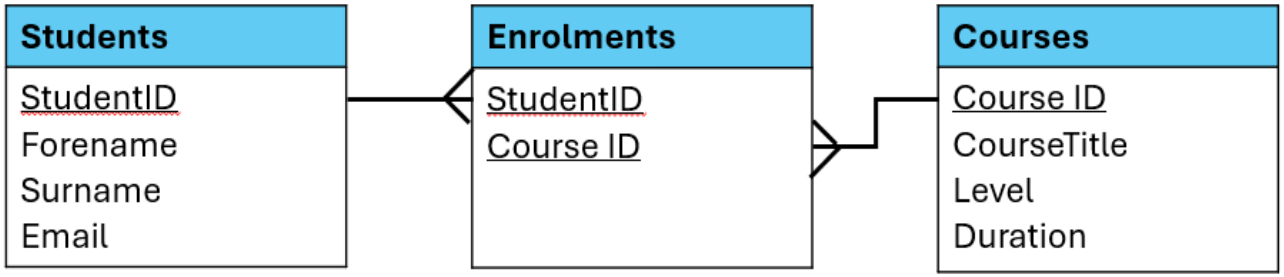
Tables created using SQL

```
CREATE TABLE Customer
(
Customer ID   INTEGER   NOT NULL
Forename      STRING    NOT NULL
Surname       STRING    NOT NULL
Email         STRING
Username      STRING    NOT NULL
Password      STRING    NOT NULL
PRIMARY KEY (Customer ID)
)
```

```
CREATE TABLE Order
(
Order ID      INTEGER     NOT NULL
Customer ID   INTEGER     NOT NULL
Product ID    INTEGER     NOT NULL
Date          DATE/TIME   NOT NULL
Quantity      INTEGER     NOT NULL
FOREIGN KEY (Customer ID) REFERENCES Customer (Customer ID)
FOREIGN KEY (Product ID) REFERENCES Product (Product ID)
PRIMARY KEY (Order ID)
)
```

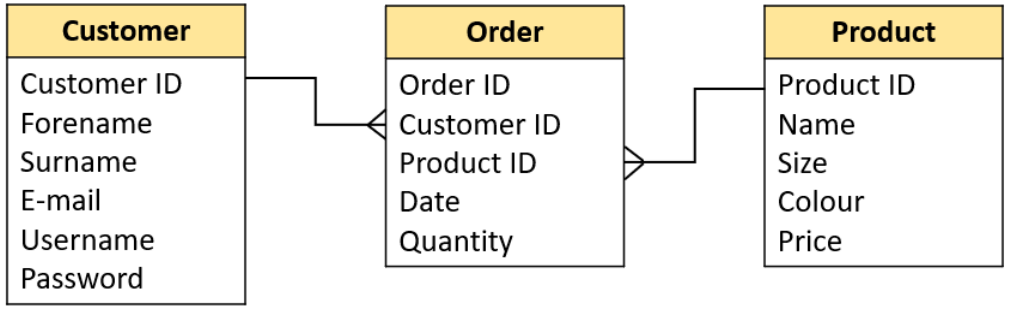
```
CREATE TABLE Product
(
Product ID    INTEGER     NOT NULL
Name          INTEGER     NOT NULL
Size          STRING
Colour        STRING
Price         FLOAT       NOT NULL
PRIMARY KEY (Product ID)
)
```

Try writing the SQL to create tables for this Relational Database:



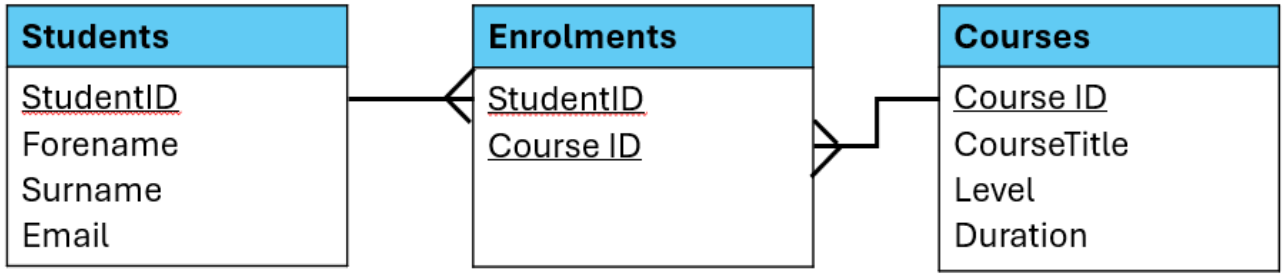
Tables created using SQL		

2.) **Inserting New Records**



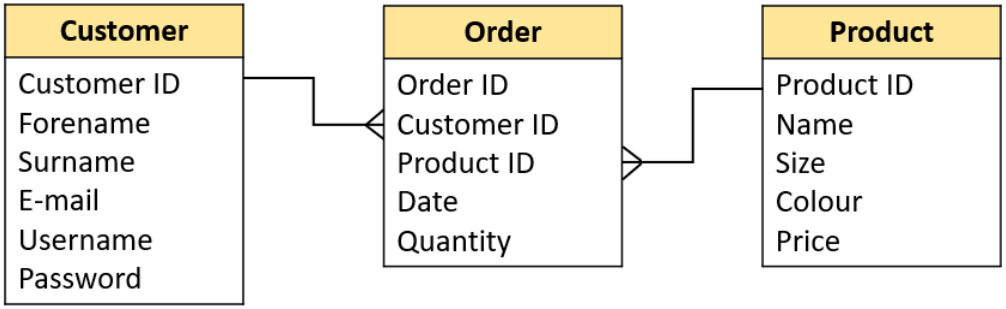
INSERT INTO Customer (Customer ID, Forename, Surname, Email, Username, Password)

VALUES (1122, "John", "Smith", JSmith@gmail.com, John.Smith, "Password123!")



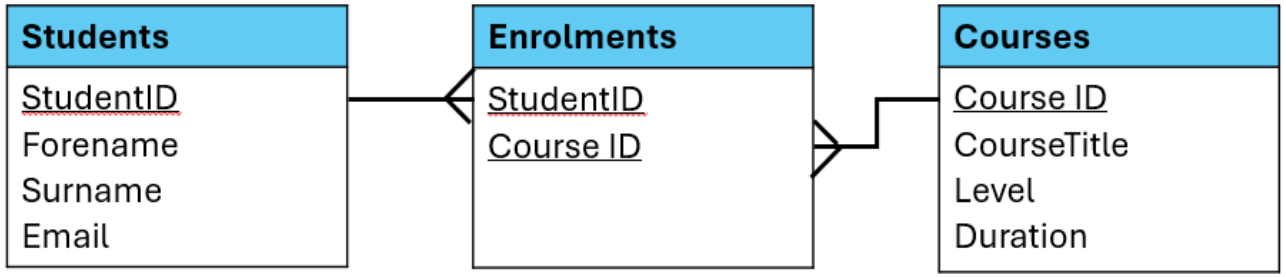
Write some SQL to insert yourself into the students table above…

3.) **Deleting Existing Records**



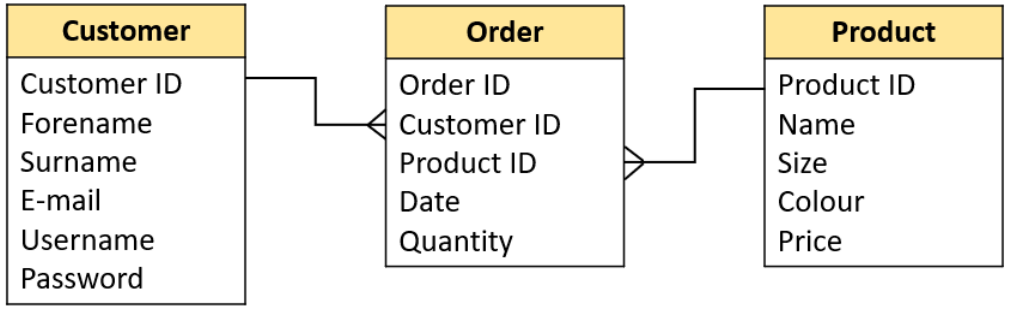
DELETE FROM Customer

WHERE Customer ID = 1122



Write some SQL to DELETE (un-enrol) yourself from T-Level Digital (course ID 1234) …

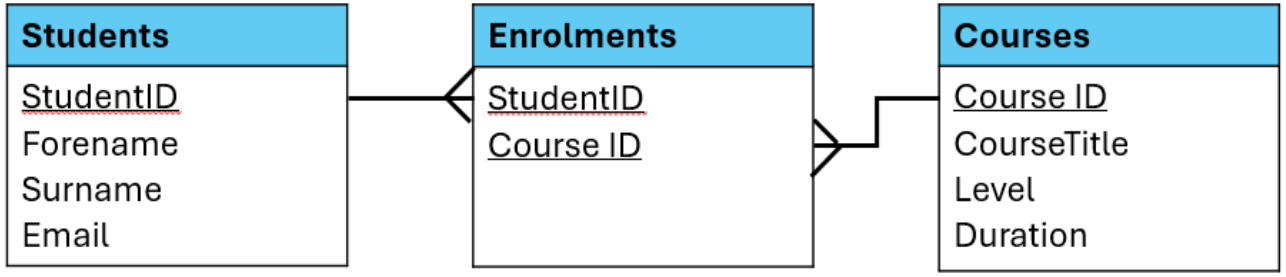
4.) **Updating Existing Records**



UPDATE Customer

SET Password = "NewPassword123!"

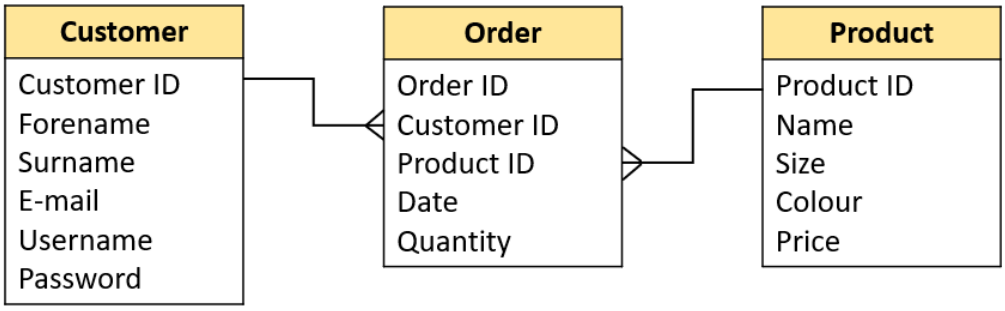
WHERE Customer ID = 1122



Write some SQL to update the course you're enrolled on to either "Hairdressing" or "Brick Laying" …

5.) **Searching/Retrieving Data (SQL Queries)**

SELECT	***List the fields you want to return***
FROM	***List the tables the fields will come from***
WHERE	***State any relationships between tables (Primary & Foreign keys)***
	+ your search criteria
ORDER BY	***Field(s) you want the results sorted by***



Example 1:

Search for contact details of all customers with the surname "Smith"

```
SELECT    Forename, Surname, Email
FROM      Customer
WHERE     Surname = "Smith"
ORDERBY   Forename
```

Example 2:

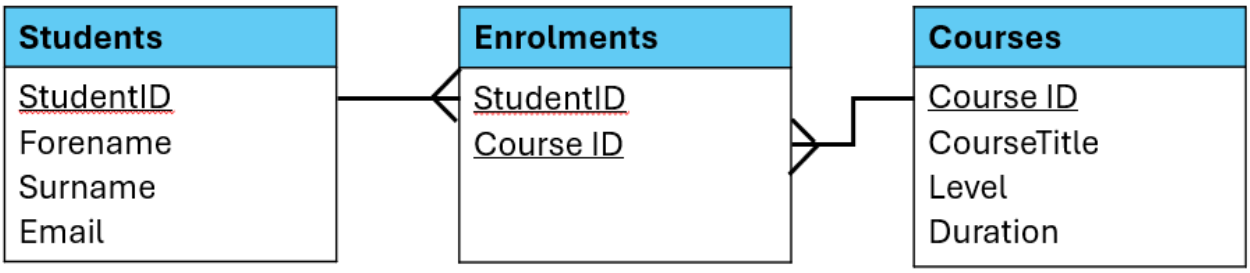
Search for all orders placed by customers called "Smith" in 2025

```
SELECT    Forename, Surname, Order ID, Product ID,
FROM      Customer, Order
WHERE     Customer.Customer ID = Orders.Customer ID AND
          AND Surname = "Smith" AND
          Date > #31/12/2024# AND Date < #01/01/2026#
                          Or
          Date BETWEEN #01/01/2025# AND #31/12/2025#

ORDERBY   Date
```

Relationship

Search Criteria



1.) Write some SQL to search for all students who take "T-Level Digital"

So that the results would look like something like this:

Student ID	Forename	Surname	Email	CourseTitle
1234	John	Smith	J.Smith@gmail.com	T-Level Digital
2054	Alice	Brown	A.Brown@yahoo.co.uk	T-Level Digital
1341	David	Morrison	DaveyM@Outlook.com	T-Level Digital

SELECT .....
FROM .....
WHERE .....
ORDERBY .....

2.) Write some SQL to search for all students with the surname "Jackson" Who are enrolled on a Level 3 Course.

The results should look like something like this:

Student ID	Forename	Surname	CourseTitle	Level
1234	Michael	Jackson	T-Level Digital	3
1423	Janet	Jackson	T-Level Digital	3
1578	Jermaine	Jackson	T-Level Digital	3

SELECT .....
FROM .....
WHERE .....
ORDERBY .....